

MLB Historical Player and Team Data Web Scraping

1. Objective

The goal of this assignment is to train students in core Web Scraping techniques and apply them to real-world big data acquisition. You are required to perform dynamic scraping of the official MLB website, handling structured data that includes time series (2003–2023) and multi-dimensional indicators (Standard + Expanded) to build a foundation for subsequent big data analysis and modeling.

2. Target URL

- MLB Official Stats Page: <https://www.mlb.com/stats/>
- Timeframe: From the year 2003 to 2023 (total of 21 seasons).

3. Detailed Requirements

You must develop a scraping program to collect data across the following four dimensions, ensuring that data for every year and every sub-tab is fully captured.

A. Individual Player Data

- **Hitting** : Path: Player -> Hitting
 - Filter Criteria: Select corresponding Year, Regular Season, MLB, All Teams, All Positions, All Players, and "Select a Split" (No special filter)
 - Data Scope: Must include all columns from both Standard and Expanded tables.
- **Pitching** : Path: Player -> Pitching
 - Filter Criteria: Select corresponding Year, Regular Season, MLB, All Teams, All Players, and "Select a Split" (No special filter)
 - Data Scope: Must include all columns from both Standard and Expanded tables.

B. Team Data

- **Team Hitting** : Path: Team -> Hitting
 - Filter Criteria: Select corresponding Year, Regular Season, MLB, and "Select a Split" (No special filter).
 - Data Scope: Must include all columns from both Standard and Expanded tables.
- **Team Pitching** : Path: Team -> Pitching

- Filter Criteria: Select corresponding Year, Regular Season, MLB, and "Select a Split" (No special filter).
- Data Scope: Must include all columns from both Standard and Expanded tables.

4. Technical Requirements

Dynamic Content Handling: Since MLB website data is mostly dynamically loaded, the use of Selenium for browser automation is recommended.

Pagination : Player data is typically spread across multiple pages. The program must automatically detect the "Next Page" button and continue scraping until all data for that year is captured.

Column Merging : As the data is split into "Standard" and "Expanded" tables, you must implement Join or Merge logic to ensure all data for the same player/team is integrated into a single row.

Exception Handling : You must account for network latency and anti-scraping mechanisms (e.g., setting an appropriate User-Agent and implementing time.sleep delays).

5. Final Deliverables

The final output should be organized into four main master tables (recommended format: CSV):

mlb_player_hitting_2003_2023.csv
 mlb_player_pitching_2003_2023.csv
 mlb_team_hitting_2003_2023.csv
 mlb_team_pitching_2003_2023.csv

6. Grading Rubric

Data Completeness: Covers all years (2003–2023) and all players/teams with no missing columns.

Program Structure: The code is modular (e.g., using functions for pagination, parsing, and file saving).

Error Handling: Implementation of automatic retry mechanisms or error logging.

Data Formatting: The output tables are clean, and column naming is correct and consistent.